

Security risk assessment report

Part 1: Select up to three hardening tools and methods to implement

To strengthen the organization's network security and prevent future breaches, three key hardening measures are recommended:

Firewall Maintenance and Port Filtering:

1. Regularly updating firewall rules and filtering ports ensures only authorized traffic enters or leaves the network. This reduces exposure to external threats and helps prevent attacks such as DDoS or unauthorized access.

Multifactor Authentication (MFA):

2. Implementing MFA adds an extra layer of protection beyond passwords. It significantly decreases the risk of account compromise, even if credentials are stolen, by requiring additional verification like a one-time code or fingerprint.

Password Policies:

3. Enforcing strong password policies—such as salting, hashing, and avoiding reuse—prevents employees from sharing or using weak passwords. This measure directly reduces the likelihood of brute force attacks and credential theft.

Together, these recommendations create a robust defense strategy that protects user data, limits unauthorized access, and ensures the network remains resilient against evolving cyber threats

Part 2: Explain your recommendations

- Firewall maintenance: Without firewall rules, malicious traffic flows freely. Port filtering and regular firewall updates ensure only authorized traffic enters or leaves the network, reducing exposure to external threats.
- Multifactor authentication (MFA): Even if a password is stolen, MFA blocks attackers by requiring an extra verification step (like OTP or fingerprint). This is one of the most effective defenses against account compromise.
- Password Policies: Employees sharing passwords and using defaults makes accounts easy targets. Strong password policies (salting, hashing, avoiding reuse) directly reduce brute force and credential theft risks.